

Vittorio Cai

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EDUCATION

Technical University of Munich (TUM) <i>M.Sc. in Management and Digital Technology</i> • Focus: Bridging business management with data science, machine learning, and software development. • Core Modules: Python and Advanced Data Science; Deep Learning(in progress); Applied Regression Analysis; Methods and Databases in Empirical Finance.	10/2025 – Present Heilbronn, Germany
Wenzhou University <i>B.A. in Financial Management</i> • Solid foundation in Financial Management, Financial Accounting, and Quantitative Methods.	09/2021 – 06/2025 Wenzhou, China

SELECTED PROJECTS

patentPATH: AI-Driven Patent Analysis <i>TUM GenAI Project</i> • Team project with Fuyao (automotive glazing) on head-up-display (HUD) windshield patents; responsible for the database and the user-facing interface. • Designed and built the PostgreSQL/pgvector database storing the patent corpus and embeddings for semantic search. • Built the web frontend where a user describes a design and receives an automated patent-overlap risk report, turning the team's analysis into output that non-technical users can act on.	05/2026 – Present
LLM News-Sentiment & Portfolio Study <i>TUM Campus Challenge</i> • Co-built a Python data pipeline processing roughly 59,000 firm-day news observations for US equities (sourcing, panel structure, merge logic). • Designed a controlled 2×2×2 factorial prompt experiment isolating event framing, chain-of-thought, and expert-role effects. • Validated economic value with Fama-French factor regressions (significant long-short alpha, $t \approx 3.6$).	11/2025 – 03/2026
Water Quality Prediction <i>Supervised ML, Data Science Course</i> • Benchmarked six models (Logistic Regression, SVM, Decision Tree, Random Forest, AdaBoost, MLP) on an imbalanced (~2:8) dataset. • Personally implemented and tuned SVM and AdaBoost; designed the cross-model evaluation (stratified k-fold CV, PR-AUC/ROC-AUC) prioritizing F1 over accuracy.	10/2025 – 01/2026

PROFESSIONAL EXPERIENCE

Ruian Municipal Bureau of Commerce <i>Finance Intern</i> • Data Analysis & Reporting: Built interactive Excel dashboards analyzing 50+ enterprises' operational performance, surfacing 2 key cost inefficiencies and turning raw data into actionable insights.	12/2024 – 01/2025 Ruian, China
Ruian Tongyan Embroidery Co., Ltd. <i>Accountant Intern</i> • Automation & Data Management: Automated wage computation from piece-rate data (saving 10+ hours/month) and built a tracking database consolidating market-price and supplier trends to support procurement decisions.	07/2024 – 09/2024 Ruian, China

SKILLS, HONORS & LANGUAGES

Programming & Data: Python (Pandas, NumPy, scikit-learn), SQL / PostgreSQL, R, SPSS, VBA.
Reporting & Tools: Interactive Excel/VBA dashboards, Matplotlib; Git/GitHub, VS Code, Jupyter / Google Colab, MS 365 (advanced Excel).
Machine Learning: PyTorch (in progress), Stable-Baselines3 (Reinforcement Learning).
Honors & Certificate: Securities Investment Competition, Provincial Third Prize (2023); Junior Accountant Cert. (China).
Languages: Mandarin (Native), English (C1), German (A1).